Aheadly

CLARITY TODAY. A BRIGHTER TOMORROW.

AI-DRIVEN APP DEVELOPMENT · PROJECT MILESTONE I

App Project Overview

AI-powered financial planning for small businesses

Aheadly turns the financial data a small business already has into clear, forward-looking cash decisions, explained in plain words.

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Aheadly at a glance

Owners of small, local businesses in Latin America have plenty of numbers but few answers. They check today's bank balance and look at spreadsheets that describe the past, so decisions about payroll, supplier payments or a new oven are made with incomplete visibility.

Aheadly is a web application that acts as a financial co-pilot for these owners. It connects once to the business's bank and QuickBooks, forecasts cash for the next eight weeks, sends an early warning in plain language when a shortfall (or idle cash) is coming, and lets the owner ask questions and test what-if scenarios in a simple chat. AI does three specific jobs: a time-series model forecasts the cash, and a large language model explains the result and turns everyday questions into scenarios.

Target user	Owners of small, local businesses in Colombia with 1 to 15 employees and no finance staff.
Persona	Maria Gómez, 45, owner of a neighborhood bakery in Bogotá.
Core promise	See a cash problem weeks before it happens, understand why, and decide with confidence.
MVP features	Connect financial accounts · Receive early-warning alert · Ask about my numbers · Test a what-if scenario
Stack	Python, Streamlit, SQLite, Prophet, Plaid, QuickBooks API, Claude or Gemini API

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1 Project Summary

1.1 Problem and target user

Who we serve. Our target user is the owner of a small, local business in Colombia, such as a bakery, café or corner shop, with 1 to 15 employees and no dedicated finance staff. The owner runs the business, handles suppliers and pays the team, and does the finances on the side.

The problem. These owners cannot see a cash shortfall coming. Supplier bills, payroll and rent often cluster in the same week, and the owner only notices when the money is already gone. Big decisions, like replacing an oven, are taken without knowing what they will do to cash in the coming weeks.

Why it matters. A single surprise shortfall means a late payment to a supplier, an overdraft fee or a missed payroll. These are direct costs, and they also damage relationships the business depends on.

How they cope today

Bank balance	Shows how much money there is today, not whether it will be enough next month.
Spreadsheets	Describe what already happened and are rarely kept up to date.
Gut feel	Combined with an accountant who reports once a month, late and at a cost the owner wants to keep low.

1.2 Solution and the role of AI

Aheadly connects to the business's bank and QuickBooks and syncs transactions every day. From that history it forecasts the cash position for the next eight weeks, detects risks (and idle cash that could be put to work), and explains them in plain language. The owner can then ask follow-up questions or test a decision before making it. Each AI job is listed below with the kind of AI used and why it fits better than a non-AI approach.

Guardrail. Every number shown to the owner comes from the forecast engine. The LLM only explains results and translates questions into scenario inputs, so it never invents figures.

1.3 MVP scope

The MVP is limited to four core features. Each one maps directly to a use case in the use case diagram (Section 3.1).

AI job	Kind of AI	Why AI, and not a fixed rule
Forecast cash flow daily, in the background	**Time-series ML model** Prophet (Python), trained on the business's own transactions	Learns each business's weekly rhythm, seasonality and recurring bills. A fixed spreadsheet rule cannot adapt to every business, and a simple average misses the week when bills cluster.
Write plain-language alerts	**LLM API** Claude or Gemini	Turns forecast numbers into a short message with the cause and two or three possible actions. A template can say *that* cash is low, but not *why* for each specific case.
Answer questions and run what-ifs	**LLM API with function calling** Claude or Gemini calling our forecast engine	Converts a question such as "What if I buy the new oven next month?" into scenario inputs, reruns the forecast and explains the result. Without AI, the owner would need finance skills to build the scenario.

Table 1: AI jobs in Aheadly, the kind of AI behind each, and why it beats a non-AI approach.

	Feature	What it does in the MVP
1	**Connect financial accounts**	Link the bank (Plaid sandbox) and QuickBooks once; transactions then sync automatically.
2	**Receive early-warning alert**	The forecast runs daily; a plain-language alert appears when a shortfall or idle cash is coming.
3	**Ask about my numbers**	Chat with Aheadly about balances, trends and upcoming bills in everyday words.
4	**Test a what-if scenario**	Model a purchase or change before deciding, with best, base and worst case.

Table 2: The four core MVP features.

Out of scope for now

- × Moving money or paying bills
- × Bookkeeping, invoicing and taxes
- × Loans or credit offers
- × Native mobile app (MVP is a Streamlit web app)
- × Multiple users or accountant access
- × Live Colombian bank feeds (MVP uses Plaid sandbox data)

Why the scope is realistic

The MVP runs on sandbox and sample data, uses a single demo business, and relies on mature, well-documented tools: Streamlit for the interface, SQLite for storage, Prophet for forecasting and a hosted LLM API. No component needs custom infrastructure, so the team can build and test it in the remaining weeks of the course.

2 Storyboard with Persona

2.1 Persona: Maria Gómez



Maria Gómez, 45

Owner of Panadería La Esquina, Bogotá

Opened her neighborhood bakery 12 years ago. She has 6 employees, serves walk-in customers and supplies two local cafés. She pays flour and dairy suppliers weekly and her team every two weeks.

"I know how to bake. I just need to know if I'll have enough cash next month."

Goals

- Never miss payroll or a supplier bill
- Know if she can afford a new oven
- Worry less about the numbers

Pain points

- Only sees today's bank balance
- Her accountant's report arrives a month late
- Bills pile up in one week and surprise her

Tech comfort: medium

- Uses a smartphone, WhatsApp and online banking daily
- QuickBooks was set up by her accountant
- Will not learn a complex finance tool

Maria is the target user from the project summary made concrete, and she is the *Business Owner* primary actor in the use case diagram.

What Maria needs from Aheadly

Her pain point	The MVP feature that answers it
Only sees today's bank balance	→ **Always-on forecast** after she connects her accounts once
Bills pile up and surprise her	→ **Early-warning alert**, weeks ahead and in plain words
Unsure if she can afford the oven	→ **Ask about my numbers** and **test a what-if scenario**

2.2 Storyboard: how Maria avoids a cash crunch

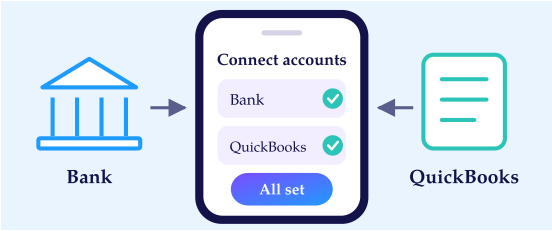
The six frames below tell the core story from trigger to outcome. The labels mark the trigger, the moments when the AI acts, and the result Maria gets.

1 Guessing — Status quo



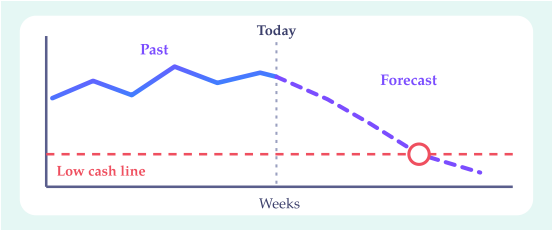
Maria runs a small bakery. She checks her bank balance to guess how things are going, and hopes for the best.

2 Connect once — Setup



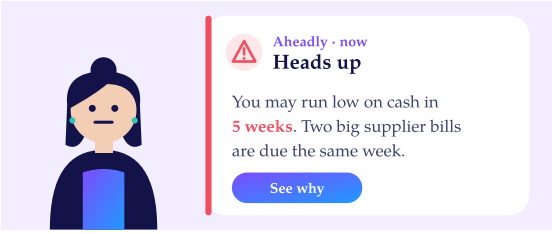
She links her bank and QuickBooks to Aheadly. It only takes a few minutes.

3 Always watching — AI forecasts



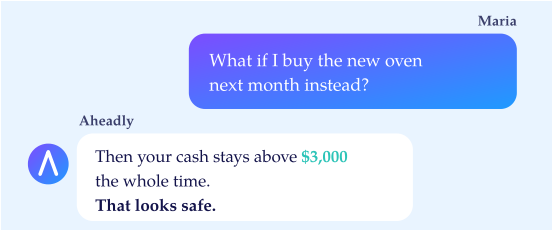
Every day Aheadly checks her money and predicts the next few weeks. Maria doesn't have to do anything.

4 Early warning — Trigger · AI alerts



Weeks before trouble hits, she gets a simple alert in plain words, with no finance terms. This is why she opens the app.

5 Ask in plain words — AI answers



She asks a follow-up question and tests a what-if before deciding, just like texting a friend.

6 Act with confidence — Outcome



Maria waits on the oven. No surprise overdraft, and she understands why. Guessing becomes planning.

3 UML Diagrams

3.1 Use case diagram

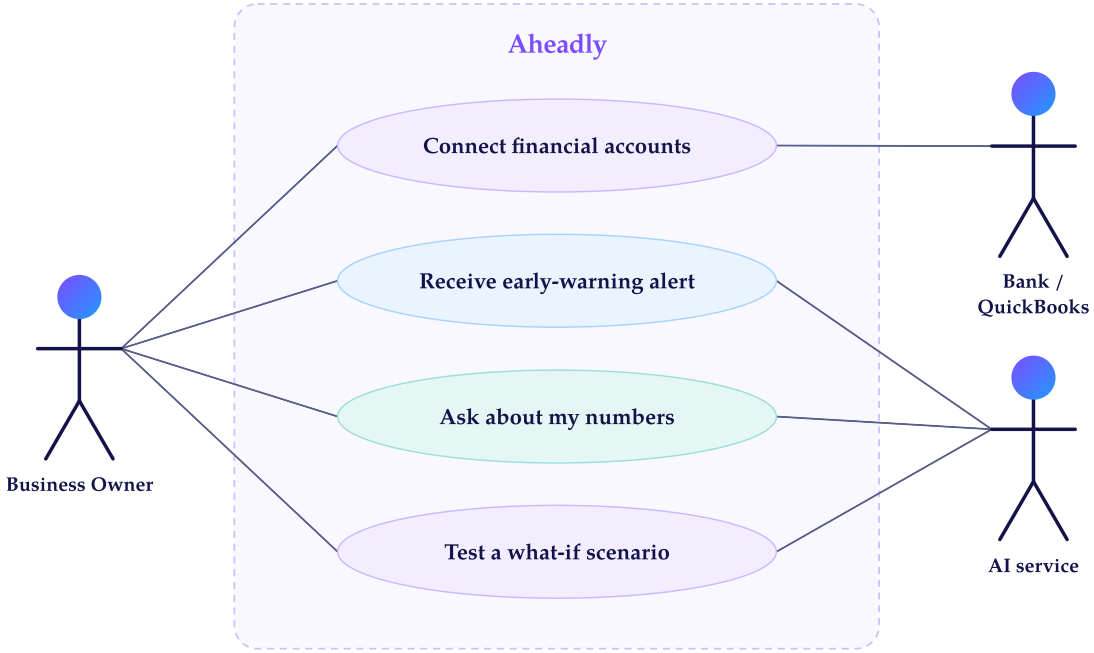


Figure 1: Use case diagram for the Aheadly MVP.

System boundary	Labeled **Aheadly**; it contains only the four MVP use cases.
Primary actor	**Business Owner** (our persona Maria), connected to all four use cases.
Secondary actors	**Bank / QuickBooks**, which supplies transaction data, and the external **AI service** (Claude or Gemini LLM API), which writes alerts, answers questions and supports what-if scenarios.
Use cases	Named as user goals (verb plus object): *Connect financial accounts*, *Receive early-warning alert*, *Ask about my numbers*, *Test a what-if scenario*.

3.2 Component diagram

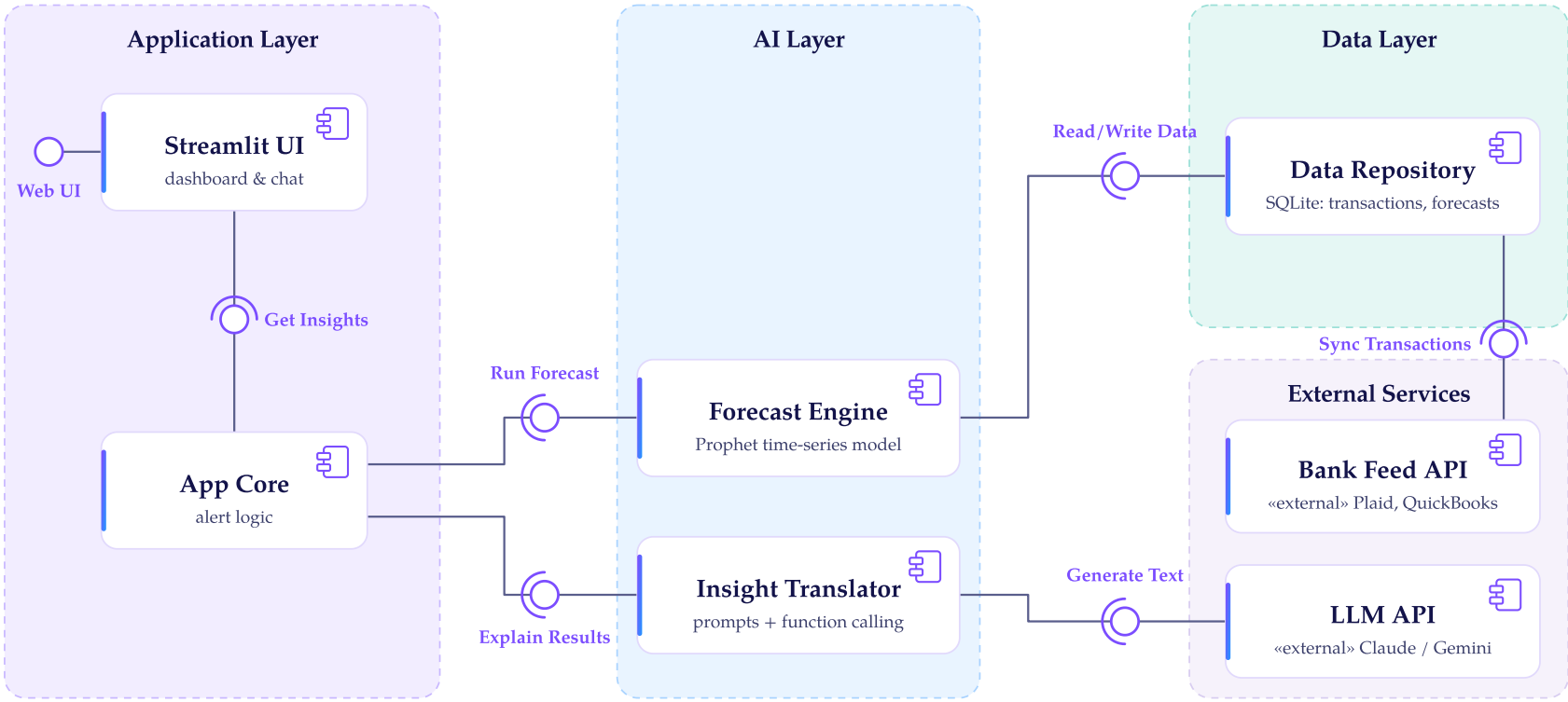


Figure 2: Component diagram grouped in three layers, with external services marked «external». Each dependency shows a provided interface (ball) and a required interface (socket).

Component	Layer	Responsibility	Technology
Streamlit UI	Application	Dashboard and chat that Maria uses	Streamlit (Python)
App Core	Application	Coordinates requests and decides when to raise an alert	Python
Forecast Engine	AI	Projects cash for the next eight weeks; reruns for what-ifs	Prophet
Insight Translator	AI	Builds prompts, explains results, parses questions via function calling	Python + LLM API
Data Repository	Data	Stores synced transactions and forecasts	SQLite
Bank Feed API	External	Supplies bank and accounting transactions	«external» Plaid, QuickBooks
LLM API	External	Generates plain-language text	«external» Claude / Gemini

Table 3: Components, their layer, responsibility and technology.

Interface	Provider (ball)	Requirer (socket)
Get Insights	App Core	Streamlit UI
Run Forecast	Forecast Engine	App Core
Explain Results	Insight Translator	App Core
Read/Write Data	Data Repository	Forecast Engine
Generate Text	LLM API	Insight Translator
Sync Transactions	Bank Feed API	Data Repository

Table 4: Interfaces: the ball marks the component that provides a service, the socket the component that needs it.

3.3 Activity diagram

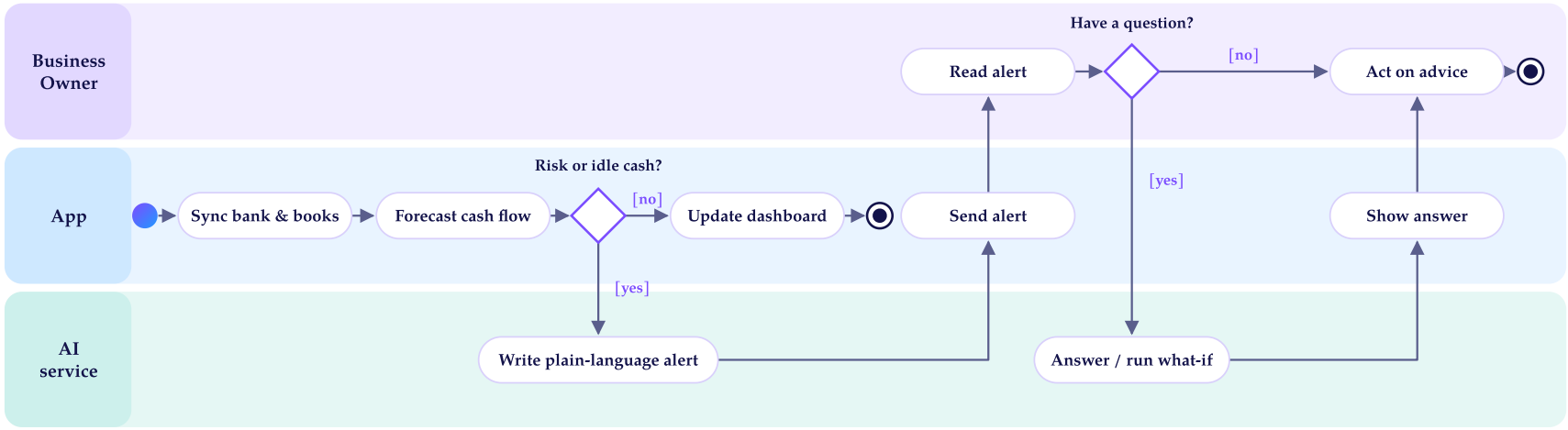


Figure 3: Activity diagram for the core use case, from data sync to decision, with swimlanes for the Business Owner, the App and the AI service.

Start and trigger
A daily sync of bank and books starts each run, followed by the cash forecast.

Visible AI steps
Write plain-language alert and *Answer / run what-if* sit in their own AI service swimlane.

Decisions and ends
Risk or idle cash? and *Have a question?* carry [yes] / [no] guards. Every path reaches a final node.

Walkthrough. The App syncs bank and accounting data and forecasts cash flow. If there is no risk or idle cash, it simply updates the dashboard and the run ends. If there is, the AI service writes a plain-language alert and the App sends it. The Business Owner reads the alert; if she has no question she acts on the advice, and if she does, the AI service answers or runs a what-if scenario and the App shows the answer before she acts. This is the same flow as storyboard frames 3 to 6.

4 Consistency Across Deliverables

All deliverables describe the same product, the same user and the same scope.

Element	Where it appears
Maria, small business owner	Target user in the project summary = persona in the storyboard = *Business Owner* primary actor in the use case and activity diagrams.
Four MVP features	The same four goals appear as the four use cases inside the Aheadly boundary.
Core story	Storyboard frames 3 to 6 (forecast, alert, question, decision) follow the activity diagram step by step.
AI and data services	Prophet, Claude or Gemini, Plaid, QuickBooks, SQLite and Streamlit named in the summary are the components and «external» services in the component diagram.

Table 5: How the summary, storyboard and UML diagrams line up.

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